$\mathcal{H}$: m-dim (m - single - particle states $c_j^\dagger$, $j = 1, \cdots, m$)

$\mathcal{A}(\otimes^n \mathcal{H})$ n - particle fermionic space ($n \leq m$)

exterior algebra

Slater determinant state (product state)

Φ : $m \times n$ matrix

$$|\Phi\rangle \in \mathcal{A}(\otimes^n \mathcal{H}) : \left(\sum \Phi_{j_1} c_j^\dagger\right)\left(\sum \Phi_{j_2} c_j^\dagger\right) \cdots \left(\sum \Phi_{j_n} c_j^\dagger\right) |0\rangle$$

$|\cdot\rangle$: $m \times n$ matrix $\longrightarrow$ n - particle state

$\Phi \longmapsto |\Phi\rangle$

$\mathrm{Im}(|\cdot\rangle)$ is not a linear subspace, since $|\Phi_1 + \Phi_2\rangle \neq |\Phi_1\rangle + |\Phi\rangle_2$, instead, it forms a mn-dimensional submanifold embedded in $\mathcal{A}(\otimes^n \mathcal{H})$, which is a $\binom{m}{n}$ - dim space.

$\hookrightarrow C_m^n$

Th 1 QR decomposition

$\forall$ $m \times n$ matrix Φ, $\exists$ $Q \in U(m)$, $m \times n$ upper triangular R, s.t.

$$\Phi = QR = Q_{jk} R_{k\ell} \qquad j, k : 1, \cdots, m \quad \ell : 1, \cdots, n$$

Cor 1

$$|\Phi\rangle = |QR\rangle = (R_{k1} Q_{jk} c_j^\dagger)(R_{k2} Q_{jk} c_j^\dagger) \cdots (R_{kn} Q_{jk} c_j^\dagger)|0\rangle$$

$$= (R_{k1} \tilde{c}_k^\dagger)(R_{k2} \tilde{c}_k^\dagger) \cdots (R_{kn} \tilde{c}_k^\dagger)|0\rangle$$

$$= \left(\prod_{i=1}^n R_{ii}\right) \tilde{c}_1^\dagger \tilde{c}_2^\dagger \cdots \tilde{c}_n^\dagger |0\rangle$$

Einstein convention

where $\tilde{c}_k^\dagger = Q_{jk} c_j^\dagger$

$\Longrightarrow |\Phi\rangle = \det(\tilde{R})\, |\tilde{Q}\rangle$,

with the property of fermion : $c_j^{\dagger 2} = 0$

$$\begin{cases} \tilde{Q} \text{ is the first } n \text{ column of } Q \\ \tilde{R} \text{ is the first } n \text{ row of } R \end{cases}$$

$|\cdot\rangle$ is not a bijection.

Th2 $\langle \bar{\Psi} | \bar{\Phi} \rangle = \det(\bar{\Psi}^\dagger \bar{\Phi})$

Proof: $\bar{\Phi} = QR \Longrightarrow \langle \bar{\Psi} | \bar{\Phi} \rangle = \langle \bar{\Psi} | QR \rangle = \langle Q^\dagger \bar{\Psi} | R \rangle$

$$\det(\bar{\Psi}^\dagger \bar{\Phi}) = \det(\bar{\Psi}^\dagger QR) = \det\left((Q^\dagger \bar{\Psi})^\dagger R\right)$$

Hence we can set $\bar{\Phi}$ to be upper triangular straightforwardly.

Without loss of generality, we set

$$\bar{\Phi} = \begin{pmatrix} \phi_{n\times n} \\ 0_{(m-n)\times n} \end{pmatrix}_{m\times n}, \quad \phi \text{ is } n\times n \text{ upper triangular, with } \det(\phi)=1$$

$$\text{Cor 1} \Longrightarrow |\bar{\Phi}\rangle = c_1^\dagger c_2^\dagger \cdots c_n^\dagger |0\rangle$$

$$\bar{\Psi} = \begin{pmatrix} \psi_{n\times n} \\ \psi'_{(m-n)\times n} \end{pmatrix}_{m\times n}$$

$$\begin{aligned}
\langle \bar{\Psi} | \bar{\Phi} \rangle &= \langle \bar{\Psi} | c_1^\dagger c_2^\dagger \cdots c_n^\dagger | 0 \rangle \\
&= \langle 0 | (\bar{\Psi}^*_{nj} c_j)(\bar{\Psi}^*_{(n-1)j} c_j) \cdots (\bar{\Psi}^*_{2j} c_j)(\bar{\Psi}^*_{1j} c_j) c_1^\dagger \cdots c_n^\dagger | 0 \rangle \\
&= \langle 0 | (\psi^*_{nj} c_j) \cdots (\psi^*_{2j} c_j)(\psi^*_{1j} c_j) c_1^\dagger \cdots c_n^\dagger | 0 \rangle \\
&= \sum_{\sigma \in S_n} \left(\prod_{j=1}^n \psi^*_{j\sigma(j)} \right) \langle 0 | c_{\sigma(n)} c_{\sigma(n-1)} \cdots c_{\sigma(1)} c_1^\dagger \cdots c_n^\dagger | 0 \rangle \\
&= \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \prod_{j=1}^n \psi^*_{j\sigma(j)} = \det(\psi^\dagger) = \det(\bar{\Psi}^\dagger \bar{\Phi}) \qquad \square
\end{aligned}$$

Th3 Define $D_{ij} = \langle \bar{\Psi} | c_i^\dagger c_j | \bar{\Phi} \rangle / \langle \bar{\Psi} | \bar{\Phi} \rangle$ if $\langle \bar{\Psi} | \bar{\Phi} \rangle \neq 0$, then

$$D^T = \bar{\Phi} (\bar{\Psi}^\dagger \bar{\Phi})^{-1} \bar{\Psi}^\dagger$$

Proof: QR decompositions: $\begin{cases} \bar{\Phi} = \tilde{U}_1 \tilde{R}_1 = (U_1 \; U_1') \begin{pmatrix} R_1 \\ 0 \end{pmatrix} = U_1 R_1 \\ \bar{\Psi} = \tilde{U}_2 \tilde{R}_2 = (U_2 \; U_2') \begin{pmatrix} R_2 \\ 0 \end{pmatrix} = U_2 R_2 \end{cases}$

$$\Longrightarrow \bar{\Psi}^\dagger \bar{\Phi} = \underset{n\times n}{R_2^\dagger} \underset{n\times m}{U_2^\dagger} \underset{m\times n}{U_1} \underset{n\times n}{R_1} \Longrightarrow (\bar{\Psi}^\dagger \bar{\Phi})^{-1} = R_1^{-1} (U_2^\dagger U_1)^{-1} (R_2^\dagger)^{-1}$$

$$\begin{aligned}
\Longrightarrow \bar{\Phi} (\bar{\Psi}^\dagger \bar{\Phi})^{-1} \bar{\Psi}^\dagger &= U_1 R_1 R_1^{-1} (U_2^\dagger U_1)^{-1} (R_2^\dagger)^{-1} R_2^\dagger U_2^\dagger \\
&= U_1 (U_2^\dagger U_1)^{-1} U_2^\dagger
\end{aligned}$$

$$\text{Cor 1} \Longrightarrow |\bar{\Phi}\rangle = \det(R_1) |U_1\rangle, \quad |\bar{\Psi}\rangle = \det(R_2) |U_2\rangle$$

$$\Longrightarrow D_{ij} = \langle u_2 | \det(R_2)\, c_i^\dagger c_j \det(R_1) | u_1 \rangle \Big/ \langle u_2 | \det(R_2) \det(R_1) | u_1 \rangle$$

$$= \langle u_2 | c_i^\dagger c_j | u_1 \rangle \Big/ \langle u_2 | u_1 \rangle$$

Hence we just need to prove

$$\frac{\langle u_2 | c_i^\dagger c_j | u_1 \rangle}{\langle u_2 | u_1 \rangle} = \left(u_1 (u_2^\dagger u_1)^{-1} u_2^\dagger \right)_{ji}$$

$$(\tilde{u}_1^\dagger D^T \tilde{u}_2)_{ij} = (\tilde{u}_1)^*_{ki} \frac{\langle u_2 | c_l^\dagger c_k | u_1 \rangle}{\langle u_2 | u_1 \rangle} (\tilde{u}_2)_{lj}$$

$$= \langle 0 | a_n a_{n-1} \cdots a_1 a_j^\dagger b_i b_1^\dagger \cdots b_{n-1}^\dagger b_n^\dagger | 0 \rangle \Big/ \det(u_2^\dagger u_1)$$

$$= (-1)^{i+j-2} \frac{\langle 0 | a_n \cdots \hat{a}_j \cdots a_1 b_1^\dagger \cdots \hat{b}_i^\dagger \cdots b_n^\dagger | 0 \rangle}{\det(u_2^\dagger u_1)}$$

$$= \begin{cases} (-1)^{i+j-2} \langle [u_2]_j | [u_1]_i \rangle = (-1)^{i+j} \dfrac{\det([u_2]_j^\dagger [u_1]_i)}{\det(u_2^\dagger u_1)} \\ 0 \qquad i > n \text{ or } j > n \end{cases}$$

where $a_j^\dagger = (\tilde{u}_2)_{ij} c_i^\dagger$, $b_j^\dagger = (\tilde{u}_1)_{ij} c_i^\dagger$

$[\text{🐱}]_i$: remove the i-th column of 🐱

$[\text{🐼}]^i$: remove the i-th row of 🐼

Further, $[u_2]_i^\dagger [u_1]_j = [u_2^\dagger u_1]^i_j$, and

$\forall$ invertible A, $(A^{-1})_{ji} = \underbrace{(-1)^{i+j} \det([A]^i_j)}_{\text{adjoint matrix}} \Big/ \det A$

$$\Longrightarrow \tilde{u}_1^\dagger D^T \tilde{u}_2 = \begin{pmatrix} (u_2^\dagger u_1)^{-1} & 0 \\ 0 & 0 \end{pmatrix} = \tilde{u}_1^\dagger \left[u_1 (u_2^\dagger u_1) u_2^\dagger \right] \tilde{u}_2 \qquad \square$$

Th4 $$\frac{\langle \Psi | c_i^\dagger c_j^\dagger c_k c_l | \Psi \rangle}{\langle \Psi | \Psi \rangle} = D_{il} D_{jk} - D_{ik} D_{jl}$$